WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

DEPARTMENT OF INFORMATION TECHNOLOGY

MAJOR PROJECT – PHASE I

PROJECT DESIGN REPORT

ON

"[INSERT PROJECT TITLE]"

Submitted in partial fulfilment of the requirements for the award of Bachelor of Technology in Information Technology

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Submitted by

Sr. No.	Student Name	Roll No.	E-mail	PRN No.

Under the Guidance of: [Guide Name]

Academic Year 2026–2027

CERTIFICATE

This is to certify that the Project Design Report entitled “[Project Title]” has been prepared by [Student Names] under the guidance of [Guide Name] as part of Major Project – Phase I, Department of Information Technology, Walchand Institute of Technology, Solapur, during academic year 2026–2027.

Guide Signature Project Coordinator Head of Department

ABSTRACT

Write a concise, project-specific summary (maximum one page) covering the domain and problem, limitations of existing solutions, proposed system, key architecture, core technical method, expected output, and present design status. Do not claim unimplemented results.

ACKNOWLEDGEMENT

Acknowledge the project guide, department, institutional support and any legitimate external contributors.

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List of Figures: Insert figure number, caption and page number.

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1.0 INTRODUCTION

1.1 Project Background

Explain the domain and real-world context in which the proposed solution will operate. Introduce the users, processes and technical environment.

1.2 Motivation and Need

Describe the specific gap or difficulty motivating the project. Explain why a technical solution is required.

1.3 Project Overview

State the project title, proposed solution, intended users and principal functions.

1.4 Aim and Objectives

State one overall aim and 4–6 specific, measurable objectives.

1.5 Scope and Limitations

Define what is included in the Phase I design, what is outside scope, and any known constraints.

2.0 LITERATURE REVIEW AND EXISTING SYSTEM ANALYSIS

2.1 Review of Recent Research

Critically review at least ten relevant research papers, preferably from the past five years, following the synopsis guidelines. For each paper identify approach, dataset or context, outcome and limitations. Cite each paper using IEEE numbering [1], [2], etc.

2.2 Comparative Literature Review

Summarise the relevant research in the following comparison table. Add rows as necessary.

Ref. / Year	Method / System	Key Findings	Limitations	Relevance
[1] / [Year]	[Approach]	[Findings]	[Limitations]	[Relevance]
[1] / [Year]	[Approach]	[Findings]	[Limitations]	[Relevance]
[1] / [Year]	[Approach]	[Findings]	[Limitations]	[Relevance]
[1] / [Year]	[Approach]	[Findings]	[Limitations]	[Relevance]

[1] / [Year]	[Approach]	[Findings]	[Limitations]	[Relevance]

2.3 Existing System Analysis

Describe existing applications, tools or processes and their workflow. Differentiate published research from currently available products where appropriate.

2.4 Research Gap and Design Implications

Identify evidence-supported limitations of the reviewed work and map each gap to a proposed design decision. Do not claim novelty without comparison.

3.0 PROBLEM DEFINITION AND REQUIREMENT ANALYSIS

3.1 Problem Statement

Express the problem clearly in 3–5 lines, retaining the approved synopsis formulation unless formally revised.

3.2 Stakeholders and Users

Identify primary users, administrators and other actors, along with their roles.

3.3 Functional Requirements

List numbered, testable system functions, e.g. FR-01: The system shall [action]. Include inputs, processing and outputs.

3.4 Non-Functional Requirements

Specify measurable security, performance, usability, reliability, scalability and privacy expectations relevant to the project.

3.5 Use Cases and Constraints

Include a labelled use-case diagram or scenarios, assumptions, dependencies and constraints.

4.0 PROPOSED SYSTEM ARCHITECTURE

4.1 Overall System Architecture

Insert a compulsory, original, project-specific architecture diagram showing actors, interface, processing modules, storage, external services and output flow. Label all arrows and interactions.

[INSERT FIGURE 4.1: OVERALL SYSTEM ARCHITECTURE DIAGRAM]

4.2 Architecture Components

Explain the responsibilities and interfaces of each major component shown in Figure 4.1.

4.3 End-to-End Data and Control Flow

Describe step-by-step how data enters, is validated, processed, stored and converted to output. Distinguish data flow from control flow.

4.4 Design Rationale

Justify the chosen architecture against the requirements, constraints and literature review findings.

5.0 DETAILED SYSTEM DESIGN

5.1 Module Identification and Responsibilities

For each module, provide purpose, inputs, operations, outputs, dependencies and error handling.

Module	Purpose	Inputs	Processing	Outputs	Dependencies
[Name]	[Purpose]	[Input]	[Logic]	[Output]	[Other modules]
[Name]	[Purpose]	[Input]	[Logic]	[Output]	[Other modules]
[Name]	[Purpose]	[Input]	[Logic]	[Output]	[Other modules]

5.2 Module Interaction Design

Insert a sequence diagram, component diagram or equivalent project-specific interaction representation.

5.3 Data / Database / Workflow Design

Choose the diagrams appropriate to the project: ER diagram, relational schema, data-flow diagram, dataset structure, workflow diagram, data pipeline or hardware data/control flow. Explain entities, attributes, relationships and validation where applicable.

5.4 API / Interface Design

Specify relevant interfaces, endpoints, request/response structure, authentication and exception cases, or hardware communication protocols. Mark non-applicable items as not applicable with a reason.

5.5 Security and Error Handling

Explain access control, input validation, sensitive data handling, logging and recovery mechanisms where relevant.

6.0 CORE TECHNICAL DESIGN AND METHODOLOGY

6.1 Technical Approach

Describe the principal algorithm, AI/ML model, recommendation engine, RAG pipeline, cloud mechanism, IoT communication, or other core methodology appropriate to the project.

6.2 Detailed Processing Steps

Provide an original algorithm, pseudocode, mathematical formulation or flowchart. Explain input, intermediate processing and final output.

6.3 Project-Specific Design Details

For AI/ML: dataset source, preprocessing, training/validation strategy and evaluation metrics. For web/mobile: backend logic, APIs, storage and authentication. For IoT/hardware: components, sensors, communication, circuit and control flow. Include only applicable sections.

6.4 Technical Contribution

Explain the specific engineering contribution, improvement or innovative element relative to reviewed approaches, with evidence or clearly labelled design intent.

7.0 USER INTERFACE AND PROTOTYPE DESIGN

7.1 User Journey and Navigation

Describe the intended user workflow and major navigation paths.

7.2 Screen Designs / Wireframes

Insert labelled wireframes or prototype screenshots for key screens and explain what each user can do. For systems without a GUI, provide an appropriate prototype or interface representation.

7.3 Prototype Status

Distinguish proposed screens, completed mock-ups and functioning prototype features.

8.0 TECHNOLOGY STACK AND IMPLEMENTATION PLANNING

8.1 Proposed Technology Stack

Give each language, framework, tool, database, API, platform and hardware item with a brief reason for selection.

Category	Proposed Technology	Version (if known)	Purpose / Justification
Programming Language			
Framework / Library			
Database / Storage			
Tools / Platform			
Hardware / External API			

8.2 Implementation Status

Provide a factual status table: completed, in progress, not started; attach evidence or links if available.

Activity / Module	Current Status	Evidence / Deliverable	Next Action
[Activity]	[Status]	[Evidence]	[Action]
[Activity]	[Status]	[Evidence]	[Action]
[Activity]	[Status]	[Evidence]	[Action]
[Activity]	[Status]	[Evidence]	[Action]

8.3 Milestones and Work Plan

Present a feasible timeline or Gantt chart for remaining development, integration, testing, deployment and documentation.

8.4 Feasibility and Risks

Summarise technical, economic, operational and schedule feasibility; identify important risks and mitigation plans.

9.0 TESTING STRATEGY AND EXPECTED OUTCOMES

9.1 Testing Plan

Specify unit, integration, system, usability, security or model evaluation tests as applicable.

Test ID	Requirement / Module	Test Input / Scenario	Expected Result	Metric / Acceptance Criterion
T-01				
T-02				

9.2 Evaluation Criteria

State the measurable criteria for determining whether the project objectives have been met.

9.3 Expected Deliverables

List expected system, prototype, model, documentation and other outputs; distinguish planned outputs from achieved results.

10.0 CONCLUSION AND FUTURE SCOPE

10.1 Design Summary

Summarise how the architecture, modules, data flow and technical method together address the problem.

10.2 Readiness for Implementation

Identify approved design elements, unresolved decisions and immediate next steps.

10.3 Future Scope

Describe reasonable extensions beyond the present scope.

REFERENCES / BIBLIOGRAPHY

Use IEEE numbered citation style, consistent with the synopsis guidelines. Include at least ten relevant recent research papers actually consulted. Example structure: [1] A. Author and B. Author, “Article title,” Journal/Conference, vol., no., pp., year, doi. Do not submit placeholders or fabricated citations.

APPENDICES (IF APPLICABLE)

Appendix A: Detailed diagrams. Appendix B: Supplementary requirements and schema. Appendix C: Prototype screenshots or design evidence. Appendix D: Work plan and supporting material.

FORMATTING AND SUBMISSION GUIDELINES

Use A4 paper; Times New Roman, 12 pt body; 14 pt bold major headings; 1.5 line spacing; justified body text; 1-inch margins; bottom-centre page numbers; numbered sections (1.0, 1.1, 1.1.1); numbered and captioned figures/tables; IEEE references. These specifications are carried

over from the supplied 2026–27 synopsis format as a recommended consistent departmental design-report format.

Design emphasis: The report must contain an original overall architecture diagram, clearly defined modules and interactions, appropriate data/database/workflow design, core technical approach, user interface or prototype, technology stack, implementation status and next milestones. Avoid generic diagrams and unsupported implementation claims.